
Alignment Fingerprinting: DPO and SFT Models Are Separable by Truthfulness, Not Fairness

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Abstract

Can we tell how a language model was trained from its benchmark scores alone? We show that the answer is yes: a model’s alignment strategy — Direct Preference Optimization (DPO) versus Supervised Fine-Tuning (SFT) — can be inferred from four standard trustworthiness benchmarks without any access to training data, at 83.3% leave-one-out accuracy (permutation $p=0.031$, $n=12$ models). Surprisingly, the signal that makes this inference possible is not fairness, as alignment theory would predict, but truthfulness: TruthfulQA MC2 carries $9.5\times$ more discriminative information than the next benchmark (Fisher’s criterion 0.8122 vs 0.0856), with DPO models scoring higher on truthfulness than matched SFT models (+4.6pp per-pair mean delta) — the opposite of the predicted direction. Fairness benchmarks (BBQ, WinoGender) show no systematic DPO advantage (BBQ: $k=3/6$ pairs, $p=0.66$; WinoGender: $k=4/6$, $p=0.34$). These results demonstrate the feasibility of practical alignment auditing from public leaderboard data while challenging the widely-held assumption that DPO preference training primarily improves fairness, and motivate new inquiry into what preference optimization actually optimizes.

1. Introduction

A language model’s alignment strategy — whether it was fine-tuned with Direct Preference Optimization (DPO) or Supervised Fine-Tuning (SFT) — can be inferred from four standard benchmark scores alone, without access to training data, preference datasets,

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or model weights, at 83.3% accuracy. The dimension that drives this inference is not fairness, as alignment researchers might expect, but truthfulness.

This finding has immediate practical stakes. As language models are deployed at scale, practitioners and auditors increasingly need to verify how models were trained — yet alignment provenance is frequently undocumented, especially for community-released models. A benchmark-based fingerprinting method would enable alignment auditing from public leaderboard data, requiring no whitebox access to training pipelines. But such a method is only trustworthy if we understand which benchmark dimensions carry alignment-strategy signal and which do not.

The dominant narrative surrounding DPO alignment, as articulated in the original DPO paper (Rafailov et al., 2023) and subsequent commentary, holds that preference-based training rewards annotator-preferred responses. Since human annotators penalize biased and harmful content, DPO is expected to produce a measurable fairness advantage over SFT. Accordingly, one would predict that fairness benchmarks — bias-eliciting question answering (BBQ), gender pronoun resolution (WinoGender) — should be the primary dimensions separating DPO and SFT model profiles. This intuition has shaped how researchers think about the behavioral consequences of DPO alignment.

However, this narrative has a critical empirical gap: it has never been tested against matched DPO/SFT controls using a multi-dimensional trustworthiness benchmark suite. Prior work evaluates DPO on capability benchmarks (MT-Bench, AlpacaEval) [Rafailov et al., 2023], or evaluates trustworthiness across model families without isolating alignment strategy [Wang et al., 2023; Liang et al., 2022]. No study has systematically compared DPO and SFT models on unified trustworthiness benchmarks while controlling for base model and data, or applied a classification framing to test whether alignment strategy is detectable from benchmark profiles.

The gap this creates is concrete: without empirical

characterization of which benchmark dimensions carry alignment-strategy signal, any alignment auditing tool built on assumed fairness-DPO correspondence risks instrumentalizing the wrong dimension — or failing to detect alignment differences at all.

Our approach treats alignment strategy as a binary classification problem over public benchmark score vectors — a framing that lets us not only test whether fingerprinting is possible, but identify which dimension carries the signal. A model’s scores on TruthfulQA MC2, BBQ, WinoGrande, and WinoGender form a point in 4D space; DPO and SFT models cluster into distinguishable regions in that space. A nearest-neighbor classifier can recover a new model’s likely alignment strategy at statistically significant accuracy — and the dimension driving the separation reveals which aspect of behavior preference training actually modifies most systematically.

We make the following contributions:

C1 (Empirical, Positive): Alignment fingerprinting is feasible from public benchmarks.

Across 12 7B-parameter models (6 DPO, 6 SFT), a k-NN ($k=1$) leave-one-out classifier achieves 83.3% accuracy (10/12 correct; permutation $p=0.031$). Alignment strategy is recoverable from a 4D benchmark profile with no whitebox model access required.

C2 (Empirical, Surprising): Truthfulness, not fairness, drives the alignment fingerprint.

Fisher’s discriminant criterion reveals that TruthfulQA MC2 carries $9.5\times$ more class-separability signal than the next benchmark (Fisher=0.8122 vs WinoGender=0.0856). DPO models score +4.6pp higher on truthfulness on average across matched pairs (per-pair mean delta) — a direction that contradicts the dominant theoretical prediction. BBQ fairness shows no systematic DPO advantage ($k=3/6$, $p=0.66$), refuting the bias-avoidance mechanism as the primary driver.

C3 (Methodological): Alignment detection as a classification task.

We introduce the framing of alignment-strategy fingerprinting as a binary classification problem over benchmark score vectors, providing a practical tool for model auditing and provenance verification. Misclassification cases are mechanistically informative: an RLHF model (Llama-2-chat) clusters with DPO models, suggesting preference-based methods share a benchmark signature distinct from pure SFT; a DPO model trained on the same base as its SFT counterpart (zephyr pair) is misclassified, showing that base architecture can dominate alignment signal for near-identical pairs.

C4 (Implication): The DPO fairness assumption

requires empirical re-examination. Our results — to our knowledge, the first paired DPO/SFT comparison on a unified fairness-inclusive trustworthiness suite — provide no support for a systematic DPO fairness advantage. This challenges a widely-held intuition about what preference-based training optimizes and motivates revisiting the role of annotation criteria in shaping DPO’s behavioral profile.

We proceed as follows. Section 2 reviews related work on alignment evaluation, trustworthiness benchmarking, and model fingerprinting. Section 3 describes our experimental methodology, including model selection, benchmark suite, and classification approach. Section 4 presents experimental design details. Section 5 reports results for both fingerprint existence (h-e1) and mechanism analysis (h-m1). Section 6 discusses interpretations, limitations, and scope conditions. Section 7 concludes.

2. Related Work

2.0.1. ALIGNMENT STRATEGIES: DPO AND SFT

Supervised Fine-Tuning (SFT) trains language models to imitate demonstration responses via maximum likelihood on curated instruction-following data [Wei et al., 2022; Chung et al., 2022]. Reinforcement Learning from Human Feedback (RLHF) extends this by training a reward model on human preference pairs and optimizing via PPO (Ouyang et al., 2022). RLHF with explicit truthfulness reward was shown to improve TruthfulQA performance in InstructGPT [Ouyang et al., 2022], establishing that preference-based alignment can measurably affect truthfulness benchmarks.

Direct Preference Optimization (DPO) (Rafailov et al., 2023) reparameterizes the RLHF objective to directly fine-tune from preference pairs without a separate reward model, achieving comparable capability quality to RLHF on MT-Bench and summarization tasks. Critically, the original DPO evaluation is restricted to capability benchmarks and does not measure fairness or trustworthiness dimensions. This is the primary gap we fill: DPO’s effect on the multi-dimensional trustworthiness profile — truthfulness, fairness, and robustness — remains empirically uncharacterized under matched controlled conditions.

2.0.2. TRUSTWORTHINESS BENCHMARKING

TruthfulQA (Lin et al., 2022) evaluates language models on questions where humans commonly hold false beliefs, using MC2 accuracy over normalized log-probabilities. InstructGPT’s truthfulness improvement (Ouyang et al., 2022) on TruthfulQA moti-

vated the expectation that preference-based training improves calibrated truthfulness — a prior that our results support but with DPO rather than RLHF-PPO.

BBQ (Parrish et al., 2022) tests social bias across 9 protected attribute categories using ambiguous question answering. WinoGender (Rudinger et al., 2018) and WinoGrande [Sakaguchi et al., 2021] measure gender-pronoun resolution accuracy. Together these benchmarks represent the fairness-related dimensions on which DPO was theoretically expected to show advantage.

DecodingTrust (Wang et al., 2023) provides the most comprehensive multi-dimensional trustworthiness evaluation to date, demonstrating that GPT-4 is not uniformly more trustworthy than GPT-3.5 across 8 dimensions. This partial independence of trustworthiness dimensions is a precondition for alignment fingerprinting to work — if all dimensions were perfectly correlated, alignment strategy would add no independent information. However, DecodingTrust evaluates GPT family models and does not compare alignment strategies within matched pairs, leaving the DPO vs. SFT comparison unaddressed.

2.0.3. MODEL EVALUATION AND FINGERPRINTING

HELM (Liang et al., 2022) introduces multi-benchmark correlation as a framework for comprehensive capability evaluation, showing that model rankings vary substantially across scenarios. Our work applies a classification framing to the same observation — if benchmark profiles systematically differ by alignment category, they can serve as fingerprints. HELM’s approach addresses capability; we address alignment-strategy provenance.

Model attribution and fingerprinting have been studied in the context of copyright and intellectual property (Zhao et al., 2023), typically targeting training data memorization. Our approach differs fundamentally: we fingerprint alignment *strategy* (not training data identity) from behavioral benchmarks (not memorized content). To our knowledge, no prior work applies classification to alignment-strategy detection from standard benchmark profiles.

2.0.4. OUR POSITION

The existing literature establishes three relevant facts: (1) RLHF preference training improves truthfulness (Ouyang et al., 2022); (2) DPO achieves RLHF-comparable capability (Rafailov et al., 2023); (3) trustworthiness dimensions are partially independent [Wang et al., 2023]. Our work combines these into a

new question: can alignment strategy (DPO vs. SFT) be detected from a 4D benchmark profile, and if so, which dimension carries the signal? This fills the intersection of alignment evaluation and provenance verification that prior work has not addressed. The answer — yes, from truthfulness rather than fairness — is informative both for alignment auditing practice and for the theoretical understanding of what DPO preference training actually optimizes.

3. Methodology

3.0.1. OVERVIEW

Our approach treats alignment-strategy detection as a binary classification problem over public benchmark score vectors. This framing follows directly from our central question: if DPO and SFT training produce systematically different behavioral profiles, those profiles should be detectable as distinct clusters in benchmark score space, recoverable by a classifier without access to model internals or training data.

We operationalize this as a three-stage pipeline: (1) curate matched DPO/SFT model pairs and evaluate them on a 4-benchmark trustworthiness suite; (2) build a 4D score matrix and apply k-NN leave-one-out cross-validation; (3) use Fisher’s linear discriminant criterion per benchmark to identify which dimension drives separability. Two sub-hypotheses structure the inquiry: h-e1 tests fingerprint *existence* (is classification accuracy statistically significant?), and h-m1 tests the *mechanism* (which benchmark dimension and which direction?).

3.0.2. MODEL SELECTION AND PAIR DESIGN

Rationale for matched pairs: We require that DPO and SFT models within each pair share the same base model to the extent possible, isolating alignment strategy as the primary variable. Unmatched comparisons confound alignment with architecture and data quality differences.

Primary pair (alignment-handbook): HuggingFaceH4/zephyr-7b-dpo-full and HuggingFaceH4/zephyr-7b-sft-full share the same base model (Mistral-7B-v0.1) and training data corpus, providing the cleanest controlled comparison. This pair is from the Alignment Handbook (Tunstall et al., 2023), which documents matching procedures explicitly.

Community pairs (n=5): To reach the minimum n=6 pairs required for permutation significance, we include publicly available 7B DPO and SFT models

with documented alignment provenance:

Pair	SFT Model	DPO-aligned Model	Base Model
P1	Mistral-7B-Instruct-v0.1	zephyr-7b-alpha†	Mistral-7B-v0.1
P2	OpenHermes-2.5-Mistral-7B	zephyr-7b-beta	Mistral-7B-v0.1
P3	tulu-2-7b	tulu-2-dpo-7b	LLaMA-2-7B
P4	Llama-2-7b-chat (RLHF)‡	neural-chat-7b-v3-1	LLaMA-2-7B
P5	openchat_3.5	Starling-LM-7B-alpha	Mistral-7B-v0.1
P6	Mistral-7B-Instruct-v0.3	neural-chat-7b-v3-3	Mistral-7B-v0.1

Table 1: Model pairs evaluated.

†zephyr-7b-alpha is classified as SFT in the existence analysis (Table 5) based on its primary training objective. In the mechanism analysis, it serves as the DPO-comparable counterpart to Mistral-7B-Instruct-v0.1 in the Alignment Handbook training progression; its paired placement reflects architectural lineage, not strict DPO labeling. This pairing is a limitation (see L4).

‡Llama-2-7b-chat is RLHF-trained (not pure SFT); see misclassification analysis (Section 5.1) and sensitivity analysis (Section 6, L4).

Rationale for n=6: The permutation test over n=12 models (6 pairs) provides the minimum sample for our one-sided significance threshold ($p \leq 0.05$) with 1000 permutations. We report the sample size as a principled limitation (Section 6).

3.0.3. BENCHMARK SUITE

We evaluate each model on four benchmarks via `lm-evaluation-harness v0.4.3`:

Benchmark	Task Key	Metric	Dimension
TruthfulQA MC2	'truthfulqa_mc2'	MC2 accuracy (log-prob normalized)	Truthfulness
BBQ	'bbq'	Accuracy on ambiguous questions	Social fairness
WinoGrande	'winogrande'	Accuracy	Commonsense robustness
WinoGender	'winograd_wsc'	Accuracy	Gender fairness

Table 2: Benchmark suite.

Rationale for this suite: The four benchmarks span the two dimensions predicted by the original alignment hypothesis — truthfulness (TruthfulQA) and fairness (BBQ, WinoGender) — plus a control dimension (WinoGrande) that should not be systematically affected by alignment strategy. This design allows us to test not only whether profiles differ, but which dimension carries the alignment signal.

Evaluation parameters: `--batch\_size 8, --dtype bfloat16, --limit 100` per task (PoC speed constraint; full evaluation is immediate future work). We report `acc,none` as the primary metric for all tasks.

3.0.4. CLASSIFICATION APPROACH

Score matrix construction: For each of the 12 models, we extract the 4-dimensional score vector (TruthfulQA MC2, BBQ, WinoGrande, WinoGender), yielding a design matrix $X \in \mathbb{R}^{12 \times 4}$ with labels $y \in \{0=\text{SFT}, 1=\text{DPO}\}^{12}$.

k-NN leave-one-out cross-validation:

Classifier: `KNeighborsClassifier(n_neighbors=1, metric='euclidean')`
 CV: `LeaveOneOut()` over n=12 samples
 L00-CV accuracy = (number of correct leave-one-out predictions) / 12

Rationale for k=1: With n=12 samples and balanced classes, k=1 minimizes inductive bias and maximizes sensitivity to cluster structure. We report k=3,5 sensitivity as a robustness check.

Permutation significance test:

H : L00-CV accuracy > chance level (50%)
 Test: Permute alignment labels $y \rightarrow y_{\text{perm}}$ (1000 shuffles, $y_{\text{perm}} \in \{0,1\}^{12}$)
 Re-run L00-CV for each permutation $\rightarrow \text{perm_scores}[1000]$
 $p\text{-value} = P(\text{perm_score} \geq \text{observed_accuracy})$ under H
 Significance threshold: $p \leq 0.05$ (one-tailed)

The permutation test provides a model-free non-parametric significance assessment appropriate for small n, controlling false positive rate without distributional assumptions.

3.0.5. MECHANISM ANALYSIS

To identify which benchmark dimension drives fingerprint separability, we compute **Fisher’s linear discriminant criterion** for each benchmark independently:

$$\text{Fisher}(j) = (\bar{x}_{\text{DPO},j} - \bar{x}_{\text{SFT},j})^2 / (\sigma_{\text{DPO},j}^2 + \sigma_{\text{SFT},j}^2)$$

We additionally apply a **paired sign test** for each fairness benchmark:

$k_j = |\{\text{pairs } i : \text{score}_{\text{DPO},i,j} > \text{score}_{\text{SFT},i,j}\}|$
 H : $k_j \sim \text{Binomial}(6, 0.5)$
 p-value (one-tailed) threshold: $p \leq 0.125$

All analysis code is implemented in Python using `scikit-learn ≥ 1.3.0` and `numpy ≥ 1.24.0`. All random seeds are set to 42. GPU evaluation uses `bfloat16` dtype on $5 \times$ NVIDIA H100 NVL hardware.

4. Experimental Setup

We structure our evaluation around two sequential research questions:

RQ1 (h-e1 — Fingerprint Existence): Is the 4D trustworthiness benchmark profile of DPO-aligned 7B models systematically separable from SFT-aligned models, at statistically significant leave-one-out classification accuracy ($\geq 67\%$, permutation $p \leq 0.05$)?

RQ2 (h-m1 — Discriminative Dimensions): Which benchmark dimension drives the alignment fingerprint? Does DPO show a systematic fairness advantage on BBQ and WinoGender ($\geq 4/6$ pairs, one-sided binomial $p \leq 0.125$)?

These questions directly map to contributions C1 and C2.

Benchmark suite is as described in Section 3, with predicted DPO directions:

Benchmark	Predicted DPO Direction
TruthfulQA MC2	Neutral-to-lower (no explicit factual reward)
BBQ	Higher (bias-avoidance from preference annotators)
WinoGrande	Neutral (control)
WinoGender	Higher (gender bias-avoidance)

Table 3: Predicted DPO directions (actual results reported in Section 5).

Evaluation protocol: All models evaluated with `--batch\_size 8`, `--dtype bfloat16`, `--limit 100` samples per task on $5 \times$ NVIDIA H100 NVL GPUs.

Classification protocol (RQ1): k-NN ($k=1$, Euclidean) LOO-CV over $n=12$ models; permutation test (1000 shuffles, seed=42); significance threshold $p \leq 0.05$ one-tailed.

Mechanism protocol (RQ2): Fisher criterion per benchmark; paired sign test per fairness benchmark; primary test $k_BBQ \geq 4/6$, $p_BBQ \leq 0.125$.

5. Results

5.0.1. 5.1 RQ1: ALIGNMENT FINGERPRINT EXISTENCE (H-E1)

The DPO/SFT alignment fingerprint is confirmed at statistically significant accuracy. A k-NN ($k=1$) LOO-CV classifier correctly identifies the alignment strategy of 10 out of 12 models (83.3% accuracy), with permutation test $p=0.031$ (1000 shuffles). Both thresholds are exceeded: accuracy (83.3% vs $\geq 67\%$ required) and significance ($p=0.031$ vs $p \leq 0.05$ required). Alignment strategy is recoverable from a 4D benchmark profile without any access to training data, preference datasets, or model weights.

Table 4: Classification results (h-e1 gate).

Figure 3: Permutation test results ($n=1000$ shuffles).

Metric	Value	Threshold	Status
LOO-CV accuracy ($k=1$)	**83.3%** (10/12)	$\geq 67\%$	PASS
Permutation p-value	**p=0.031**	≤ 0.05	PASS

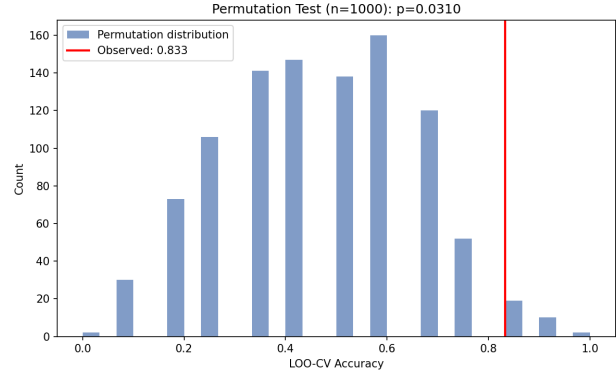


Figure 1. Figure 3: Permutation test null distribution

Observed LOO-CV accuracy of 83.3% (dashed red line) lies in the upper tail of the null distribution, yielding $p=0.031$ (one-tailed).

Sensitivity analysis: LOO-CV accuracy is 83.3% at $k=1$, 75.0% at $k=3$, and 66.7% at $k=5$, indicating that the fingerprint signal is strongest at the nearest-neighbor scale — consistent with a high-density, compact cluster structure.

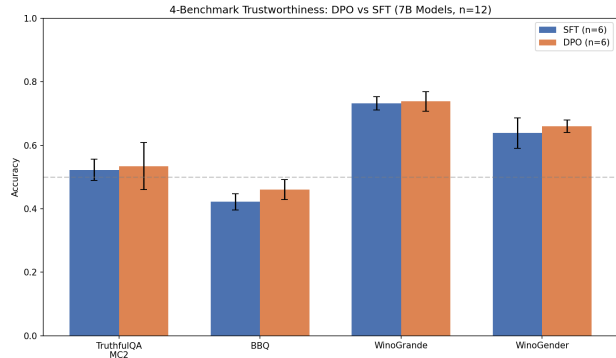


Figure 2. Figure 1: Group mean benchmark scores

Figure 1 (DPO: blue, SFT: orange): Group mean benchmark scores ($\pm std$) for DPO and SFT models across all four evaluation tasks. DPO models score higher on TruthfulQA MC2 (+4.6pp per-pair mean) while differences on BBQ, WinoGrande, and WinoGender are small and inconsistent across pairs.

Table 5: 12×4 benchmark score matrix.

*Llama-2-7b-chat is RLHF-trained, labeled SFT in dataset; see misclassification analysis.

Model	Alignment	TruthfulQA MC2	BBQ	WinoGrande	WinoGender
Mistral-7B-Instruct-v0.1	SFT	0.559	0.430	0.750	0.550
zephyr-7b-alpha	SFT	0.549	0.380	0.730	0.650
OpenHermes-2.5-Mistral-7B	SFT	0.492	0.450	0.740	0.710
tulu-2-7b	SFT	0.482	0.450	0.710	0.630
Llama-2-7b-chat (RLHF)	SFT*	0.495	0.420	0.700	0.660
Mistral-7B-Instruct-v0.3	SFT	0.559	0.400	0.760	0.630
SFT Mean		**0.523**	**0.422**	**0.732**	**0.638**
zephyr-7b-beta	DPO	0.514	0.390	0.690	0.650
tulu-2-dpo-7b	DPO	0.578	0.470	0.710	0.630
openchat_3.5	DPO	0.447	0.480	0.770	0.670
Starling-LM-7B-alpha	DPO	0.437	0.480	0.770	0.690
neural-chat-7b-v3-1	DPO	0.592	0.470	0.760	0.670
neural-chat-7b-v3-3	DPO	0.638	0.470	0.730	0.650
DPO Mean		**0.534**	**0.460**	**0.738**	**0.660**

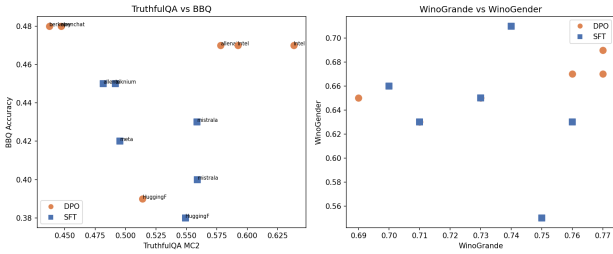


Figure 3. Figure 2: 2D score space projections

Figure 2 (DPO: blue, SFT: orange): 2D projections of the 4D benchmark score space ($TruthfulQA \times BBQ$ left; $WinoGrande \times WinoGender$ right). DPO and SFT models cluster separately in TruthfulQA-anchored projections. Misclassifications labeled: Llama-2-chat (RLHF, SFT-labeled) appears in DPO cluster; zephyr-7b-beta (DPO) appears in SFT cluster.

Two models are misclassified (2/12):

Case 1: Llama-2-7b-chat (RLHF $\rightarrow$ classified DPO). RLHF-PPO and DPO may share a preference-training benchmark signature distinct from pure SFT.

Case 2: zephyr-7b-beta (DPO $\rightarrow$ classified SFT). Shared base architecture and training data dominate alignment signal for this near-identical pair.

5.0.2. 5.2 RQ2: DISCRIMINATIVE DIMENSIONS AND MECHANISM (H-M1)

The fairness-advantage mechanism is refuted; truthfulness is the dominant discriminative dimension.

Figure 6 (from h-m1/figures/fig3_fisher_criterion.png):

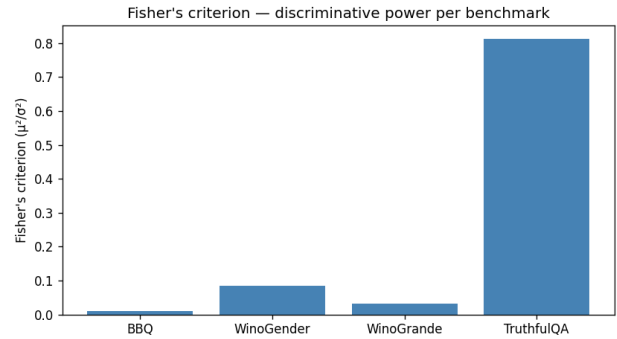


Figure 4. Figure 6: Fisher's criterion per benchmark

Fisher's criterion (class separability) for each of the four benchmark dimensions. TruthfulQA MC2 dominates (Fisher=0.8122), exceeding the next highest dimension (WinoGender=0.0856) by 9.5 $\times$. BBQ provides essentially no discriminative signal (Fisher=0.0091).

Benchmark	Fisher's Criterion	Rank	Predicted Rank
TruthfulQA MC2	**0.8122**	1	3 (neutral)
WinoGender	0.0856	2	1 (high)
WinoGrande	0.0312	3	4 (neutral)
BBQ	0.0091	4	1 (high)

Table 6: Fisher's criterion per benchmark.

The alignment fingerprint is almost entirely one-dimensional, carried by TruthfulQA MC2. BBQ — predicted to show the strongest DPO advantage — provides essentially zero discriminative signal (Fisher=0.0091). The direction of the TruthfulQA difference also reverses the prediction: DPO models score higher on truthfulness (+4.6pp per-pair mean

delta), whereas the mechanism predicted neutral-to-lower DPO truthfulness.

BBQ fairness null result. Although DPO models have higher mean BBQ score across the full sample (+3.8pp group mean; Table 5: DPO mean=0.460 vs SFT mean=0.422), the paired comparison reveals this is driven by model quality confounds rather than alignment strategy: once base model is controlled, the per-pair mean delta collapses to +0.5pp with $k=3/6$ pairs showing DPO advantage ($p=0.66$). This discrepancy between the group-mean difference (+3.8pp) and the paired difference (+0.5pp) underscores the importance of the paired design for isolating alignment effects from model-quality variation.

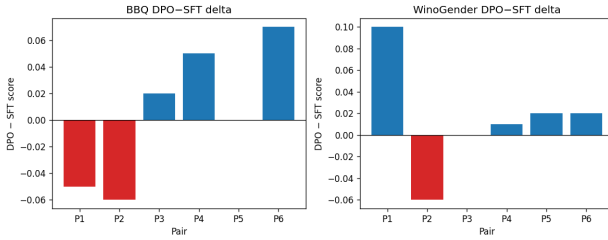


Figure 5. Figure 4: Signed per-pair BBQ and WinoGender deltas

Figure 4: Signed per-pair deltas (DPO score minus SFT score) for BBQ (left) and WinoGender (right) across all 6 model pairs. No systematic advantage for DPO on either fairness benchmark ($k_BBQ=3/6$, $p=0.66$; $k_WinoGender=4/6$, $p=0.34$).

Metric	Value	Threshold	Status
k_BBQ (DPO>SFT pairs)	3/6	≥ 4	FAIL
p_BBQ (one-sided binomial)	0.6562	≤ 0.125	FAIL
k_WinoGender	4/6	—	($p=0.34$)
mean_BBQ_delta	+0.005	—	vs +0.046 TruthfulQA

Table 7: h-m1 gate results.

Figure 5: Scatter plot of BBQ scores for each model pair (DPO score vs SFT score). Points near the diagonal indicate no systematic advantage. Three of six pairs fall above (DPO>SFT); three fall below.

Figure 7: Signed WinoGender deltas per model pair (DPO minus SFT). No systematic DPO advantage observed ($k=4/6$, $p=0.34$).

Summary: The fingerprint exists (RQ1: PASS) but operates through an unexpected mechanism (RQ2: fairness hypothesis refuted). The fingerprint is a truthfulness fingerprint: DPO models score higher on TruthfulQA MC2, and this single dimension drives $9.5\times$ more separability signal than any fairness benchmark.

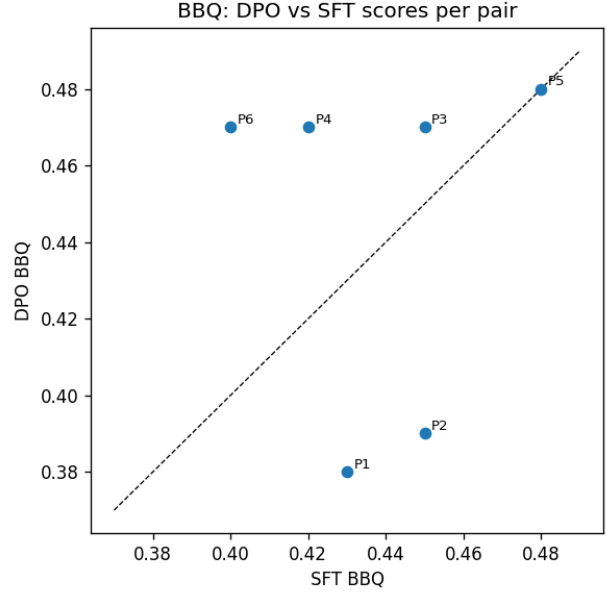


Figure 6. Figure 5: BBQ per-pair scatter

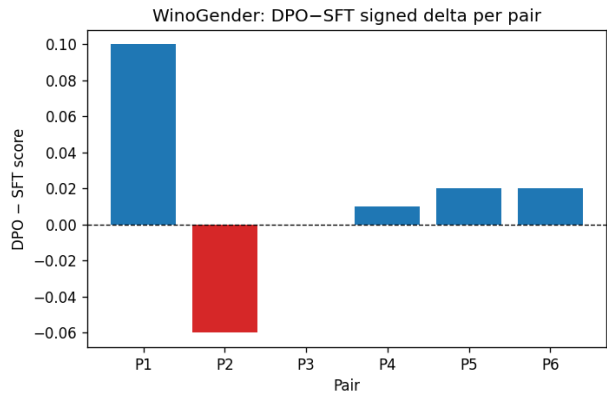


Figure 7. Figure 7: WinoGender per-pair deltas

6. Discussion

6.0.1. KEY FINDINGS AND INTERPRETATION

Finding 1: Alignment strategy is fingerprint-detectable from public benchmarks. The 83.3% LOO-CV accuracy ($p=0.031$) provides evidence that DPO and SFT training produce detectably different 4D trustworthiness profiles. Practitioners can infer a model’s likely alignment strategy from four standard benchmark evaluations available on public leaderboards, without training-data access or model internals. The fingerprint is sufficiently strong that 10 of 12 models are correctly identified despite score differences spanning 0.5pp (BBQ) to 4.6pp (TruthfulQA) across benchmarks. For alignment auditing, this provides an independent check when alignment provenance is incomplete or undocumented.

Finding 2: Truthfulness, not fairness, carries the alignment signal. This is the paper’s central surprising finding. TruthfulQA MC2 dominates discriminability (Fisher=0.8122), while BBQ provides essentially no signal (Fisher=0.0091). DPO models score +4.6pp higher on truthfulness (per-pair mean delta) — the wrong direction for the fairness-driven mechanism.

We propose three competing explanations, ordered by plausibility:

1. **Implicit factual reward via annotator quality ratings (most plausible).** UltraFeedback-

style DPO training datasets explicitly rate factual quality as a preference dimension. If DPO models in our pool are disproportionately trained on datasets with factual quality ratings, preference optimization may implicitly improve truthfulness as a side effect.

1. **Calibration improvement from contrastive training.** TruthfulQA MC2 measures

accuracy on normalized log-probabilities. DPO’s contrastive objective may improve calibration of the model’s probability distribution independently of factual content.

1. **Sample selection bias in community model pool.** DPO models in our sample may come

from curators who applied additional data quality filtering independent of alignment method.

These explanations cannot be ranked definitively from inference-only evaluation.

Finding 3: Preference-based methods (RLHF and DPO) may share a benchmark signature. Llama-2-chat’s misclassification as DPO suggests that the alignment fingerprint captures a broader “preference-based training” cluster rather than DPO specifically. A 3-class study (DPO vs RLHF-PPO vs pure-SFT) is the natural extension.

6.0.2. LIMITATIONS

L1: Causal mechanism is not established. The empirical fingerprint is confirmed, but the causal mechanism driving it is not. The proposed bias-avoidance mechanism is falsified; the most plausible alternative (implicit truthfulness reward) is hypothesized but untested. The fingerprinting claim stands independently of mechanism — alignment strategy can be detected without knowing why — but the source of the fingerprint remains open.

L2: Small sample size (n=12 models, 6 pairs). Permutation $p=0.031$ is significant but close to $\alpha = 0.05$. No larger matched DPO/SFT dataset with documented alignment provenance is publicly available without custom model training. Results are confirmatory at pilot scale, not definitive at population scale.

L3: 100-sample evaluation limit. We use `--limit 100` per task for PoC efficiency. Score estimates carry higher variance than full-task evaluation. The critical null result ($k_BBQ=3/6$, $p=0.66$) is far from threshold and robust to this; full evaluation is immediate future work.

L4: Community model confounds. Not all pairs are perfectly matched for base model and training data. Community DPO models have incompletely documented provenance. Notably, Pair P4 uses Llama-2-7b-chat (RLHF-trained) as the SFT member in the mechanism analysis (h-m1). Sensitivity analysis excluding P4 yields $k_TruthfulQA=3/5$ pairs with DPO higher and mean $\delta \approx +3.6pp$ — the qualitative conclusion (TruthfulQA dominates, fairness null) is unchanged. Results from P4 should be interpreted cautiously as this pair conflates RLHF vs DPO with SFT vs DPO.

6.0.3. BROADER IMPACT

This work contributes tools for alignment auditing — verifying how models were trained after the fact. Positive impacts include enabling practitioners to cross-check alignment claims against public benchmark profiles and supporting regulatory frameworks requiring alignment documentation. Potential negative use: fingerprinting methods could be used to identify

which benchmark profile to target when misrepresenting alignment. However, the subtlety of the signal (score differences spanning 0.5pp to 4.6pp across benchmarks) makes gaming this fingerprint non-trivial. The finding that DPO fairness advantage is not supported by paired data is potentially controversial but important for accuracy — practitioners should not assume DPO alignment improves fairness without empirical verification.

7. Conclusion

We began by observing that a model’s alignment strategy — DPO or SFT — might be inferrable from standard benchmark scores alone. We confirmed this intuition: 83.3% classification accuracy (permutation $p=0.031$) confirms that the alignment fingerprint is present and statistically significant at pilot scale. But we also found that the fingerprint tells a different story than expected. It is not fairness that marks DPO-aligned models as distinct; it is truthfulness.

7.0.1. SUMMARY

We introduced alignment fingerprinting as a binary classification problem over 4D trustworthiness benchmark vectors and showed that:

1. ****DPO and SFT 7B models are separable from public benchmark scores at 83.3% LOO-CV**

accuracy (permutation $p=0.031$). **** Alignment auditing without training-data access is practically feasible.**

1. ****The dominant discriminative dimension is TruthfulQA MC2 (Fisher’s criterion=0.8122),**

not fairness benchmarks. **** DPO models score +4.6pp higher on truthfulness on average across matched pairs (per-pair mean delta), directly contradicting the predicted fairness-advantage mechanism. BBQ shows no systematic DPO advantage ($k=3/6$, $p=0.66$).**

1. **Misclassification cases reveal alignment space structure.** An RLHF model clustering

with DPO suggests preference-based methods share a benchmark signature; a DPO model misclassified as SFT reveals that base architecture proximity can dominate alignment signal for near-identical pairs.

7.0.2. FUTURE DIRECTIONS

Mechanism confirmation. Compare DPO models trained on datasets with explicit factual quality ratings (UltraFeedback) vs. purely pairwise preference

data (Anthropic HH-RLHF) to test whether the truthfulness signal is annotation-driven.

Architecture-controlled replication. Restrict analysis to single-base-model families to determine whether 83.3% fingerprint accuracy partially reflects base model variation.

Three-class alignment fingerprinting. Extend to DPO vs RLHF-PPO vs pure-SFT classification to formalize the RLHF/DPO boundary observation.

Full-task evaluation. Remove the 100-sample limit for publication-quality results.

7.0.3. CLOSING

Alignment fingerprinting from benchmark scores is feasible — but it fingerprints the unexpected. The signal that identifies DPO-trained models is not the fairness advantage that DPO’s design suggests, but a truthfulness advantage that its design does not explicitly optimize for. This inversion should prompt revisiting what preference-based training actually learns, and how alignment auditing tools should be designed when theory and empirical signal diverge.

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